

Retail Customer Segmentation - Portfolio Project

A customer segmentation and cross-sell analysis, built using SQL, Excel, and Power BI, applied to a real customer transaction dataset.

Approach

I applied RFM segmentation, a simple and interpretable method suited to an initial customer analysis a business stakeholder can act on directly. The method was applied to real e-commerce transactions from the Olist Brazilian E-Commerce dataset (Kaggle). Illustrative brokerage-style attributes - product adoption, risk profile, and acquisition channel - were added on top of the real segments. The pipeline was built in SQL (SQLite), covering data cleaning, customer identity resolution, and segmentation logic. Excel was used for validation: manual spot-checks against the SQL output, a chi-square test on the adoption pattern, and the cross-sell opportunity calculation. Power BI holds the final three-page dashboard used to present the results.

Method

- Identified customers using `customer_unique_id`, not the `per-order customer_id` field, to avoid counting each order as a separate customer. The raw data contains 99,441 order-level records but only 96,096 unique customers.
- Cleaned three real tables and excluded canceled or unavailable orders, reducing 99,441 orders to 98,207 valid transactions. Of the 96,096 unique customers, 94,983 had at least one valid completed order; the remaining 1,113 were excluded, as none of their orders were valid.
- Calculated Recency and Monetary scores using SQL window functions (`NTILE(4)`), splitting customers into four equal-sized groups by each measure. Frequency required a different approach: 96.9 percent of customers ordered exactly once, so a standard four-part split would not have shown any real difference between groups. A simple repeat-versus-one-time flag was used instead.
- Built six segments using CASE logic, then checked each one against its own underlying quartile scores. This check found that 18,081 customers, or 19 percent of the base, had been placed in a general catch-all group despite scoring in the top half (quartile 3 or 4) on recency or monetary value. This was corrected by adding a separate "Steady / Moderate Value" segment,

defined as customers scoring quartile 3 or higher on both recency and monetary value.

- Calculated the cross-sell opportunity for each segment in Excel and ran a chi-square test to confirm the differences in adoption were unlikely to be due to chance ($p < 0.001$).

Key Finding

The six segments span a wide range of value and engagement. New / Low Value and Low Value together account for exactly half of all customers (47,492 of 94,983) but show minimal spend and low product adoption, making them a lower priority for cross-sell in the near term. Among the remaining segments, Champions has the highest adoption rate per customer, at 41.6 percent, and the highest average spend, but is limited in scale, at 510 customers. This analysis weighs total opportunity size above adoption rate as the primary criterion, on the assumption that a larger, moderately engaged segment produces more total value than a small, highly engaged one. This assumption has not been tested against real data and is presented here as a judgment, not a proven result.

Under this criterion, Steady / Moderate Value offers the strongest balance: 18,081 customers, an illustrative opportunity of approximately 544,000 BRL, and 30.1 percent existing adoption. At Risk shows a larger opportunity figure, at approximately 844,000 BRL, but its low recency score makes re-engagement uncertain before any cross-sell attempt.

Recommendation

I recommend prioritizing the Steady / Moderate Value segment for margin cross-sell. Champions, at 41.6 percent adoption, and High-Value One-Time, at 39.8 percent adoption, are strong secondary candidates given their higher conversion likelihood per customer, despite smaller overall scale. New / Low Value and Low Value are not recommended for near-term targeting given their limited spend and adoption. Any real targeting would need to exclude customers with a conservative risk profile, in line with KYC and suitability requirements, regardless of segment.

Limitations

Segment membership, recency, and monetary values are based on real transaction data. Product adoption, risk profile, and acquisition channel are illustrative: rule-based, and tied to real monetary and frequency values, but not discovered patterns. This recommendation rests on two untested assumptions: a 20 percent conversion

rate, and the choice to weigh total opportunity above adoption rate. Both would require validation against real data before informing an actual business decision.

Dataset: Olist Brazilian E-Commerce Public Dataset, Kaggle: kaggle.com/datasets/olistbr/brazilian-ecommerce

Project files: The SQL pipeline, Power BI dashboard, and Excel workbook are available on request.

Tools: Built using Excel and Power BI. The internship job description also references Superset, which was not used in this project.